MAT 3103: Computational Statistics and Probability

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Chapter 1 [Data representation & Descriptive Statistics]

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Data Representation

Data Representation: We can often make a large and complicated set of data more compact and easier to understand. Statistical data can be presented by

i) Tabulation method
ii) Graphs and diagrams method

i) **Tabulation method:**

Frequency Distribution: A tabular arrangement of data by classes together with the corresponding number of items in each class is called a frequency distribution or frequency table. It is used to represent the value of different levels of quantitative variable. Example of frequency distribution is as-

Table: Tally marks for grouping the length of 40 laurel leaves

Class Interval of length	Tally	Frequency
118-128	///	3
128-138	/𝍸/ //	7
138-148	/𝍸/ 𝍸/// ///	13
148-158	/𝍸/ ////	9
158-168	///𝍸	5
168-178	///	3
Total		40

Terms Associated with Frequency Distributions:

- Class size or width - the differences between lower and upper class limits.
- Cumulative frequencies are the cumulative totals of successive frequencies of a frequency distribution.
- Class mark or midpoint - the average of class limits.

Length	Frequency	Midpoint	Cumulative frequency
118-128	3	123	3
128-138	7	133	10
138-148	13	143	23
148-158	9	153	32
158-168	5	163	37
168-178	3	173	40
Total	40		

- Find the number of leaves which length is less than 158mm.
  Ans: There are 3+7+13+9 = 32 leaves whose length are less than 158mm.
- Find the percent of leaves which length is above and 148 mm.
  Ans: There are $\frac{9+5+3}{40} \times 100 = 42.5\%$ leaves whose values are above and 148mm.

ii) Graphs and diagrams method:

Different graphs and diagrams are-
i) Bar diagram ii) Pie diagram
iii) Histogram iv) Scatter diagram.

Diagrammatic representation of data:
Bar diagram and pie diagram are generally used to represent the value of qualitative variable diagrammatically.

Bar diagram: Bar diagrams are simple diagrams that are made up of a number of rectangular bars of equal widths whose heights are proportional to the quantities or frequencies they represent.

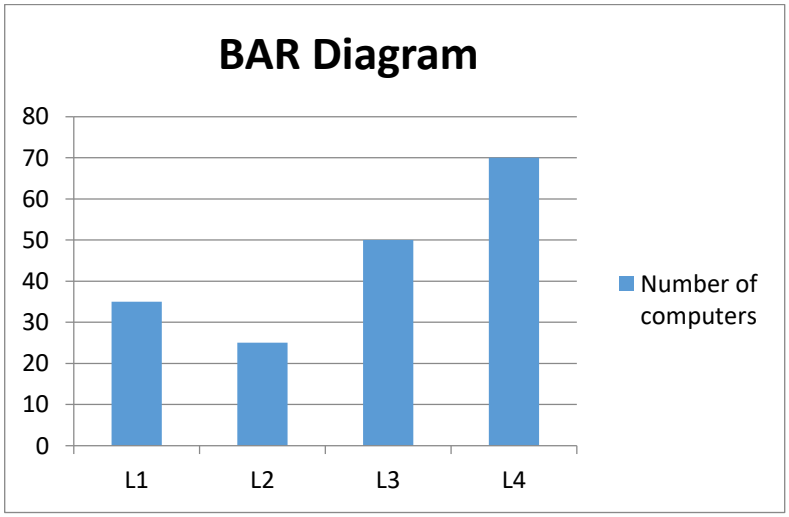
Pie diagram: Pie diagrams can be defined as a circle drawn to represent the totality of a given data. The circle is also divided into sectors with each sector proportional to the components of the variable it represents.
Pie diagram is very useful in drawing comparison among the various components or between a part and the whole. Both diagrams are used to represent value of different levels of qualitative variable.

Example: The following are the number of computers available in different laboratories.

Laboratory	Number of computers
L1	35
L2	25
L3	50
L4	70
Total	180

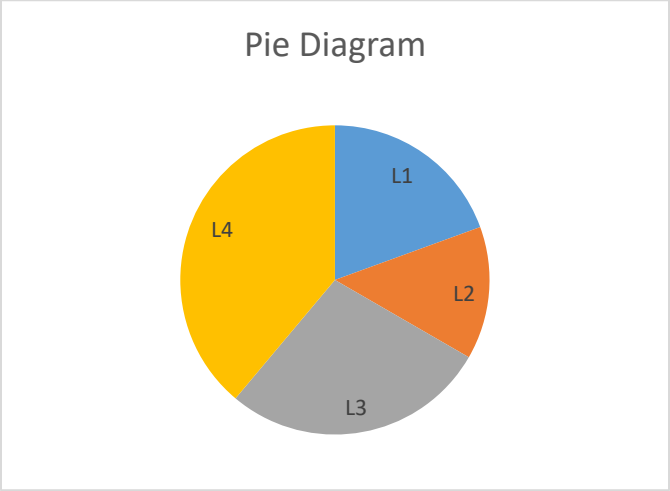
a) Draw a bar diagram of the data.
b) Represent the data by a pie diagram.

Solution:
(a)



(b)We have calculated below the various angles of pie-diagram:

Laboratory	Number of computers	Angels=$\frac{x\times 360^\circ}{Total}$
L1	35	70
L2	25	50
L3	50	100
L4	70	140
Total	180	360

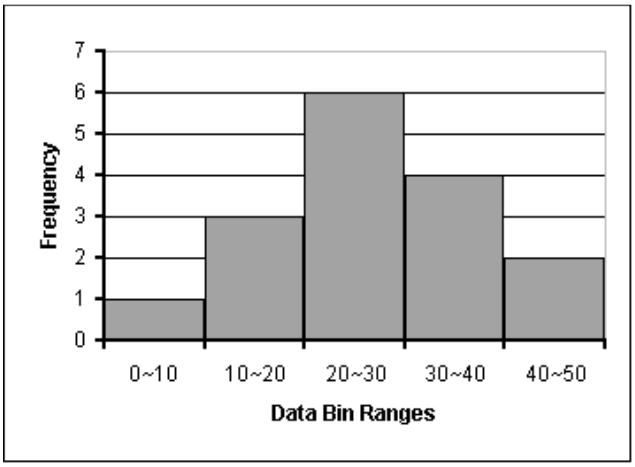


Histogram: Consists of set of rectangles having: (a) bases on a horizontal axis with centers at the class marks and length equal to the class interval sizes, and (b) areas proportional to the class frequencies. It is the graphical representation of continuous classes of frequency distribution.

Example: A frequency distribution the changing the size of the bin is given as. Draw a histogram using the given data set.

Class Interval	**Frequency**
0-10	1
10-20	3
20-30	6
30-40	4
40-50	2
Total	16

Solution: Histogram of the changing the size of the bin



Example: The following is the distribution of weights (in kg) of 50 persons:

Weight (in kgs)	*50-55*	*55-60*	*60-65*	*65-70*	*70-75*	*75-80*	*80-85*	*85-90*	*Total*
Number of persons	*12*	*8*	*5*	*4*	*5*	*7*	*6*	*3*	*50*

Draw a histogram for the above data.

Solution: We represent the class limits along the X-axis on a suitable scale and the frequencies along the Y-axis on a suitable scale.
Since the scale on the X-axis starts at 50, a kink (break) is indicated near the origin to signify that the graph is drawn to scale beginning at 50, and not at the origin.
Do yourself in the class.

- **Difference between Bar diagram and histogram:**
  a. Bar diagram is used to represent the qualitative variable and histogram is used to represent quantitative variable.
  b. Bar diagram is one dimensional and histogram is two dimensional.

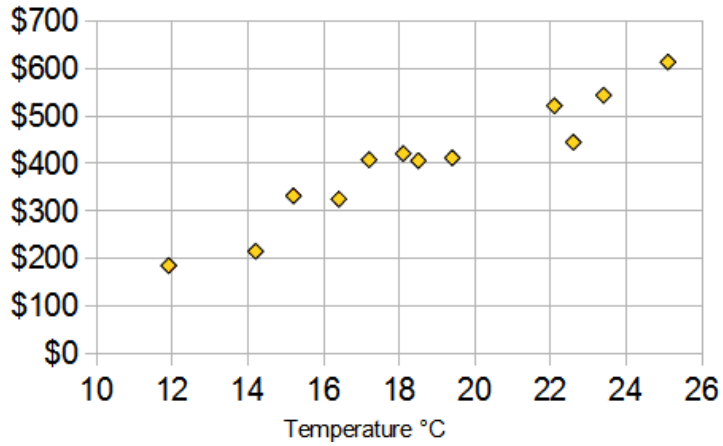
Scatter diagram: A scatter (XY) Plot has points that show the relationship between two sets of data.

In this example, each dot shows one person's weight versus their height.
To construct a **Scatter plot**, Label the x- and y- axis. Choose a range that includes the maximums and minimums from the given data.

Example: The local ice cream shop keeps track of how much ice cream they sell versus the noon temperature on that day. Here are their figures for the last 12 days:

Ice Cream Sales vs Temperature	
Temperature °C (X)	**Ice Cream Sales (Y)**
14.2°	215
16.4°	325
11.9°	185
15.2°	332
18.5°	406
22.1°	522
19.4°	412
25.1°	614
23.4°	544
18.1°	421
22.6°	445
17.2°	408

Solution: Here is the same data as a Scatter Plot:



It is now easy to see that **warmer weather leads to more sales**, but the relationship is not perfect.

MATLAB code for the related topic:

Diagrams:

bar produces vertical bar chart, and can be used as bar(y) or bar(x,y) – the first form uses 1:length(y) as the values for x, which are the bar locations.

```
>> x = [1 2 4 5 9];
>> y = 20-(5-x).^2;
>> bar(x,y)
>> title('Bar Chart')
```

Other bar charts can be produced using an optional style argument (bar(x,y,'style')), where style is one of:

• 'grouped' - Produces a bar chart where values in each column of y are grouped together, but appear in different colors.

• 'hist' - produces a bar chart with no space between bars.

```
>> bar(1:3,[1 2 3;2 3 4;3 4 5],'grouped')
>> bar(1:3,[1 2 3;2 3 4;3 4 5],'hist')
```

pie can be used to produce a pie chart

```
>> pie([.7 .2 .1],[.1 0 0],{'Stocks','Bonds','Cash'})
>> title('Asset allocation')
```

Sample questions related to this part:

1. *Write down the differences between histogram and bar diagram?*
2. *Which of the graphs and diagrams are used for presenting qualitative data and which are used for quantitative data?*

3. *The following are the number of calls received by person in different days:*

Days	:	*1*	*2*	*3*	*4*	*5*	*Total*
Number of calls :		*7*	*8*	*5*	*15*	*10*	*45*

Represent the number of calls of different days by bar diagram and by pie diagram. Also read the data set using MATLAB and write MATLAB code for bar diagram and pie diagram.

4. *Frequency distribution of the resting pulse rate in healthy volunteers (N = 63). Represent the data by histogram. Also read the data set using MATLAB and write MATLAB code for histogram.*

Pulse/min	*No. of volunteers*
60-65	*2*
65-70	*7*
70-75	*11*
75-80	*15*
80-85	*10*
85-90	*9*
90-95	*6*
95-100	*3*
Total	*63*

5. *The following are the number of e. mails received in different days by different organizations :*

Days (x)	:	*5*	*8*	*3*	*10*	*15*
No. of mails received (y)	:	*54*	*65*	*12*	*128*	*158*

Draw scatter diagram of the data.

Descriptive Statistics

Measures of Central Tendency

Measures of Central Tendency: Measures of central tendency are statistical constants which enable us to comprehend in a single effort the significance of the whole. A number which represents the entire list of numbers. A value which describes some attributes of the population. Its helps us to condense data in a single value. A score that indicates where the center of the distribution tends to be located.

There are different types of measures of central tendency, each has its own advantages and disadvantages. These are-

i. **Mean**
   a) **Arithmetic mean**
   b) **Geometric mean**
   c) **Harmonic mean**
ii. **Median**
iii. **Mode**

a) Arithmetic mean: It is generally known as average. To find the mean, add up all the numbers and divide by the number of numbers. It is denoted as $\bar{x}$ or AM.

For ungrouped data set,

Let $x_1, x_2, \ldots\ldots\ldots\ldots\ldots\ldots\ldots, x_n$ are n variates, then, the arithmetic mean is defined by

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

Example: The marks scored by 6 students in quiz in statistics is given below.

x: 5, 8, 10, 12, 13, 6

Calculate arithmetic mean for given data set.

Solution: Arithmetic mean,

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

$$= \frac{5+8+12+10+13+6}{6} = \frac{54}{6} = 9$$

For Frequency distribution,

Let $x_1, x_2, \ldots\ldots\ldots\ldots\ldots\ldots\ldots, x_n$ are n variates with frequencies $f_1, f_2 f_3, \ldots\ldots, f_n$ then, the arithmetic mean is defined by

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i f_i}{n}$$

Where,

x = mid value

f = frequency, n = total frequency

Example: Calculate the arithmetic mean of mark of the following distribution.

Marks	10-20	20-30	30-40	40-50	50-60	Total
No. of students	3	5	9	3	2	22

Solution: The calculation is shown in the table below:

Marks	10-20	20-30	30-40	40-50	50-60	Total
No. of students (f_i)	3	5	9	3	2	22
Mid value (x_i)	15	25	35	45	55	
$x_i f_i$	45	125	315	135	110	730

$$\text{Arithmetic mean} = \bar{x} = \sum_{i=1}^{n} \frac{x_i f_i}{n}$$

$$\bar{x} = \frac{730}{22} = 33.18$$

Advantages of the arithmetic mean:

- It is rigidly defined.
- It is easy to calculate and simple to follow.
- It is based on all the observations.
- It is determined for almost every kind of data.

Disadvantages of the arithmetic mean:

- The arithmetic mean is highly affected by extreme values.
- It cannot average the ratios and percentages properly.
- It is not an appropriate average for highly skewed distributions.
- It cannot be computed accurately if any item is missing.

b) **Geometric mean:** It is sometimes preferred for averaging ratios of two variables: rates of population growth, rates of interest, and rates of depreciation. The geometric mean of a set of n values of a variable is the nth root of their product.

For ungrouped data,

Let, a variable x assumes n values $x_1, x_2, \ldots\ldots\ldots, x_n$, then its geometric mean, denoted by $\bar{x}_G$ or GM is

$$\bar{x}_G = (x_1 \cdot x_2 \cdot \ldots \cdot x_n)^{\frac{1}{n}} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}$$

$$= \sqrt[n]{x_1 . x_2 . \ldots\ldots\ldots\ldots\ldots . x_n}$$

$$= Antilog\left(\frac{1}{n}\sum_{i=1}^{n} log x_i\right)$$

For a frequency distribution,

Let $x_1, x_2, \dots\dots\dots\dots\dots, x_n$ are n variates with frequencies $f_1, f_2 f_3, \dots\dots, f_n$ then, the geometric mean is defined by

$$\bar{x}_G = \left(\prod_{i=1}^{r} x_i^{f_i}\right)^{\frac{1}{n}}, \qquad \text{where} \quad n = \sum_{i=1}^{r} f_i$$

$$= \sqrt[n]{(x_1^{f_1}.x_2^{f_2} \dots\dots\dots.. x_n^{f_n})}$$

$$= Antilog(\frac{1}{n}\sum_{i=1}^{n} f_i log x_i)$$

Example:

What is the geometric mean of 4, 9, 9, and 2?

Solution:

$$GM = \sqrt[n]{x_1.x_2.\dots\dots\dots\dots\dots.x_n} = \sqrt[4]{4 \times 9 \times 9 \times 2} = 5.045$$

Example: Find the geometric mean of the following values:
x: 15, 12, 13, 19, 10.

Solution:

Given data, 15, 12, 13, 19, 10

The calculation is shown in the table below:

x	$log\ x_i$
15	1.1761
12	1.0792
13	1.1139
19	1.2788
10	1.0000
Total	5.648

$$GM = Antilog\left(\frac{1}{n}\sum_{i=1}^{n} log x_i\right) = Antilog\left(\frac{5.648}{5}\right) = 13.48$$

Example: Find the geometric mean of the following data:

Marks	0-10	10-20	20-30	30-40	40-50	Total
No. of students	4	8	10	6	7	35

Solution: The calculation is shown in the table below:

Class	Mid x_i	log x_i	f_i	$f_i \log x_i$
0 - 10	5	0.6990	4	2.7960
10 - 20	15	1.1761	8	9.4088
20 - 30	25	1.3969	10	13.9790
30 - 40	35	1.5441	6	9.2646
40 - 50	45	1.6532	7	11.5724
Total			35	47.0208

$$GM = Antilog(\frac{1}{n}\sum_{i=1}^{n} f_i log x_i) = Antilog\left(\frac{47.0208}{35}\right) = 22.06$$

Advantages of the geometric mean:

- The geometric mean is rigidly defined.
- The geometric mean is directly based on all the observations.
- Generally, the presence of a few extremely small or large values has no considerable effect on geometric mean.

Disadvantages of the geometric mean:

- It is difficult to compute.
- If a single value of a variable is zero, then the geometric mean becomes zero, irrespective of the magnitudes of the other values.
- It may be imaginary if some values are negative (generally, use of geometric mean is restricted to positive values).

c) **Harmonic Mean:** It is also one of the different kinds of average and it is applied for the situations when you have desired rates of average. It is calculated by dividing the number of observations by the sum of reciprocal of the observation.

For ungrouped data,

Let, a variable x assumes n values $x_1, x_2, \ldots\ldots\ldots\ldots\ldots\ldots, x_n$, then its harmonic mean, denoted by $\bar{x}_H$ or HM is

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

For a frequency distribution,

Let $x_1, x_2, \ldots\ldots\ldots\ldots\ldots\ldots, x_n$ are n variates with frequencies $f_1, f_2 f_3, \ldots\ldots, f_n$ then, the harmonic mean is defined by

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{f_i}{x_i}}$$

Example: Find the harmonic mean of the following data

x: 8, 9, 6, 11, 10, 5.

Solution:

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

$$= \frac{6}{\frac{1}{8}+\frac{1}{9}+\frac{1}{6}+\frac{1}{11}+\frac{1}{10}+\frac{1}{5}}$$

$$= 7.560$$

Example: Find the harmonic Mean for the following data

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency	3	12	8	7	15	26	5	3	1

Solution: The calculation is shown in the table below:

Class	Frequency, f_i	Mid value, x_i	$\frac{1}{x_i}$	$f_i \frac{1}{x_i}$
10-20	3	15	0.067	0.200
20-30	12	25	0.040	0.480
30-40	8	35	0.029	0.229
40-50	7	45	0.022	0.156
50-60	15	55	0.018	0.273
60-70	26	65	0.015	0.400
70-80	5	75	0.013	0.067
80-90	3	85	0.012	0.035
90-100	1	95	0.011	0.011
	80			1.849

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{f_i}{x_i}}$$

$$= \frac{80}{1.849}$$

$$= 43.27$$

Advantages of the harmonic mean:

- It is based on all observations.
- It is capable of algebraic treatment.
- It is an appropriate average for averaging ratios and rates.
- It does not give much weight to the large items

Disadvantages of the geometric mean:

- Its calculation is difficult.
- It gives high weight-age to the small items.
- It cannot be calculated if any one of the items is zero.
- It is usually a value which does not exist in the given data.

Exercise:

a) A goes to school at 20 km per hour and returns back to home at 30 km per hour. What is the average speed?

b) A fellow travels from city A to city B. For the first 15 miles, he drove at the constant speed of 20 miles per hour. Then he (instantaneously) increased his speed and, for the next 5 miles, kept it at 30 miles per hour. Find the average speed of the motion.

❖ **Show by an example that, AM ≥ GM ≥HM**

Solution: Let a data set
x : 2,3,4,5,6

Find Arithmetic mean for data**,**

$$AM = \frac{2+3+4+5+6}{5}$$
$$AM = \frac{20}{5}$$
$$AM = 4$$

Find Geometric mean for data

$$GM = \sqrt[5]{2 \times 3 \times 4 \times 5 \times 6}$$
$$GM = \sqrt[5]{720}$$
$$GM = 3.7279$$

Find Harmonic mean for data

$$HM = \frac{5}{\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}}$$
$$HM = 3.4483$$

In this case, AM > GM > HM

If data set is as

x: 5, 5, 5, 5, 5

Then, AM=GM=HM=5

So, AM ≥ GM ≥HM [Showed]

ii) **Median:** A value which divides a data set into equal two parts. To find the median, place all the numbers in order and select the middle number.

For ungrouped data,

- ***Make an array***

$Me =$ The value of $\frac{1}{2}(n+1)th\ number\ observation$ $\quad [n\ odd]$

$=$ The value of $\frac{1}{2}\left(\frac{n}{2}th\ number\ observation + (\frac{n}{2}+1)th\ number\ observation\right)$ $\quad [n\ even]$

Exercise:

Find the Median of the given data set.

x: 8, 4, 3, 3, 5, 7, 1

Solution:

Array , *x:* 1 3 3 4 5 7 8

n= 7 (an odd number)

$$Me = \text{The value of } \frac{1}{2}(n+1)th\ number\ observation$$

$$= \text{The value of } \frac{1}{2}(7+1)th\ number\ observation$$

$$= \text{The value of } 4th\ number\ observation$$

$$= 4$$

Exercise:

Find the Median of the given data set.

x: 8, 4, 3, 3, 5, 7, 1,6

Solution:

Array , *x:* 1 3 3 4 5 6 7 8

n= 8 (an even number)

$$\text{Me} = \text{The value of } \frac{1}{2}\left(\frac{n}{2}th\ number\ observation + \left(\frac{n}{2}+1\right)th\ number\ observation\right)$$

$$= \text{The value of } \frac{1}{2}\left(\frac{8}{2}th\ number\ observation + \left(\frac{8}{2}+1\right)th\ number\ observation\right)$$

$$= \text{The value of } \frac{1}{2}(4th\ number\ observation + 5th\ number\ observation)$$

$$= \frac{4+5}{2}$$

$$= 4.5$$

For a frequency distribution,

$$Me = L + \frac{\frac{n}{2} - c}{f} \times h$$

Where,

$L = Lower\ limit\ of\ median\ class$

$h = Size\ of\ median\ class$

$f = Frequency\ corresponding\ to\ the\ median\ class$

$c = Cumulative\ frequency\ of\ preceding\ class\ of\ median\ class$

$n = Sum\ of\ the\ frequencies$

Example: Following distribution of statistics scores of 20 students

Scores	1-2	2-3	3-4	4-5	5-6	Total
No. of students	1	3	8	6	2	20

Find the value of median of scores.

Solution: The calculation is shown in the table below:

Class	Frequency, f	Cumulative frequency
1-2	1	1
2-3	3	4
3-4	8	12
4-5	6	18
5-6	2	20
Total	20	

$$Me = L + \frac{\frac{n}{2} - c}{f} \times h$$

$$= 3 + \frac{\frac{20}{2} - 4}{8} \times 1$$

$$= 3.75$$

Advantages of Median:

- It is very simple to understand and easy to calculate.
- Median lies at the middle part of the series and hence it is not affected by the extreme values.
- It is a special average used in qualitative phenomena like intelligence or beauty which are not quantified but ranks are given.
- In grouped frequency distribution it can be graphically located by drawing ogives.

Disadvantages of Median:

- In simple series, the item values have to be arranged. If the series contains large number of items, then the process becomes tedious.
- It is a less representative average because it does not depend on all the items in the series.
- It is not capable of further algebraic treatment.
- It is not rigidly defined.

iii) Mode:

The mode is the number which appears most often. The most frequently occurring score. There can be more than one mode in a data set. If the two values are tied for being the most common values in the set, the data set can be said to be *bimodal*, whereas if three values are tied, the set is *trimodal*, and so on.

Example: The data set is given as
X: 2, 3, 3, 4, 6.
Find the value of mode.
Answer: Mo= 3.

The data set is given as
X: 2, 3, 3, 4, 6, 6.
Find the value of mode.
Answer: Mo= 3and 6.

The data set is given as
X: 2, 3, 3, 4, 4, 4, 6.
Find the value of mode.
Answer: Mo= 4.

The data set is given as
X: 2, 3, 4, 6.
Find the value of mode.
Answer: Mo= No mode.

For frequency distribution,

$$Mo = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

where,

$L = Lower\ limit\ of\ modal\ class$

$h = Size\ of\ modal\ class$

$f_m = Frequency\ corresponding\ to\ the\ modal\ class$

$f_1 = Frequency\ preceding\ to\ the\ modal\ class$

$f_2 = Frequency\ proceeding\ to\ the\ modal\ class$

Example: The frequency distribution is given as, Calculate the value of mode.

Class	Frequency
0-10	2
10-20	10
20-30	5
30-40	6
40-50	4
Total	27

Answer:

$$Mo = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$
$$= 10 + \frac{10 - 2}{2 \times 10 - 2 - 5} \times 6$$
$$= 13.43$$

Advantages of mode:

- Good with nominal data.
- Bimodal distribution might verify clinical observations.
- Easy to compute and understand.
- The score exists in the data set.

Disadvantages of mode:

- Ignore most of the information in a distribution.
- Small samples may not have a mode
- More than one mode might exist.

Measures of dispersion

Measures of dispersion: The measures of central tendency are not adequate to describe data. Two data sets can have the same mean but they can be entirely different. Thus to describe data, one needs to know the extent of variability. This is given by the measures of dispersion. Dispersion means the scatteredness of data from central value. Another way of examining single variable data is to look at how the data is spread out, or dispersed about the mean. It gives us the information of average deviation from central value.

Consider two sets of data to understand dispersion:

X_1: 23, 24, 25, 26, 27; n_1=5; mean= 25

X_2: 5, 10, 16, 28, 66; n_2=5; mean= 25

We easily observe that the mean for both the data is 25. But the set are not identical as the variates are either scattered widely or closely packed.

Measures of dispersion:

i) Mean deviation
ii) Variance
iii) Standard deviation
iv) Coefficient of Variation (C.V)

i) Mean deviation: The mean deviation or average deviation is the arithmetic mean of the absolute deviations and is denoted by $D_{\bar{x}}$ or MD.

Mean Deviation for ungrouped data,

$$\text{Mean deviation , } MD = \frac{\sum_{i=1}^{n}|x_i-\bar{x}|}{n},$$

Example: Calculate the mean deviation of the following distribution.

x: 9, 3, 8, 8, 9, 8, 9, 18

$$\bar{x} = 9$$

$$MD = \frac{|9-9| + |3-9| + \cdots \ldots . + |18-9|}{8}$$

$$= 2.25$$

Mean Deviation for frequency distribution,

If the data is grouped in a frequency table, the expression of the mean deviation is:

$$\text{Mean deviation, } MD = \frac{\sum_{i=1}^{n} f_i |x_i - \bar{x}|}{n}$$

Example: Calculate the mean deviation of the following distribution:

Class Interval	10-15	15-20	20-25	25-30	30-35	Total
Frequency	3	5	7	4	2	21

Solution:

Class Interval	x_i	f_i	$x_i\ f_i$	$\lvert x_i - \bar{x} \rvert$	$f_i \lvert x_i - \bar{x} \rvert$
10-15	12.5	3	37.5	9.286	27.858
15-20	17.5	5	87.5	4.286	21.43
20-25	22.5	7	157.5	0.714	4.998
25-30	27.5	4	110	5.714	22.856
30-35	32.5	2	65	10.714	21.428
Total		21	457.5		98.57

$$\bar{x} = 21.786$$

$$MD = \frac{98.57}{21} = 4.69$$

Advantages of Mean Deviation:

- Mean deviation is rigidly defined and its value is precise and definite.
- Mean deviation is readily understood. It is the average of the deviations from a measure of central tendency.
- It is based on all the observations.
- It is not affected very much by the value of the extreme items.

Disadvantages of Mean Deviation:

- Mean deviation ignores the algebraic signs of the deviations, and as such, it is not capable of further mathematical treatment.
- Mean deviation is not a very accurate measure of dispersion.

ii) Variance: The variance is defined as the square of the mean. The deviations are taken from arithmetic mean of the data.

Variance for ungrouped data,

$$\sigma^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n}$$

Variance for grouped data,

$$\sigma^2 = \sum_{i=1}^{n} \frac{f_i(x_i - \bar{x})^2}{n}$$

iii) Standard deviation: The standard deviation is defined as the positive square root of the mean.

Standard deviation for ungrouped data,

$$S.D = \sigma = S = \sqrt{\sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n}}$$

$$= \sqrt{Variance}$$

Standard deviation for frequency distribution,

$$S.D = \sigma = S = \sqrt{\sum_{i=1}^{n} \frac{f_i(x_i - \bar{x})^2}{n}}$$

iv) Coefficient of Variation:

The most important of all the relative measure of dispersion is the coefficient of variation. The coefficient of variation (*C.V*) is defined as:

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

Thus *C.V* is the value of *S* when $\bar{x}$ is assumed equal to 100. It is a pure number and the unit of observations is not mentioned with its value. It is written in percentage form like 20% or 25%. When its value is 20%, it means that when the mean of the observations is assumed equal to 100, their standard deviation will be 20. The *C.V* is used to compare the dispersion in different sets of data particularly the data which differ in their means or differ in the units of measurement.

Example: Calculate variance, standard deviation and coefficient of variation for the following sample data.

x: 2, 4, 8, 6, 10, 12.

Solution:

X	$(x_i - \bar{x})^2$
2	$(2-7)^2=25$
4	$(4-7)^2=9$
8	$(8-7)^2=1$
6	$(6-7)^2=1$
10	$(10-7)^2=9$
12	$(12-7)^2=25$
$\sum x_i=42$	$\sum(x_i - \bar{x})^2=70$

$$\bar{x} = \frac{\sum x}{n} = \frac{42}{6} = 7$$

$$\sigma^2 = \frac{\sum(x-\bar{x})^2}{n} = \frac{70}{6} = 11.67$$

$$\sigma = \sqrt{11.67} = 3.42$$

Coefficient of Variation

$$CV = \frac{\sigma}{\bar{x}} \times 100 = 48.86\%$$

Example: Calculate coefficient of standard deviation and coefficient of variation from the following distribution of marks:

Marks	1-3	3-5	5-7	7-9	9-11	Total
No. of students	12	15	11	8	4	50

Solution:

Marks	f_i	x_i	$f_i x_i$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
1−3	12	2	24	9	108
3−5	15	4	60	1	15
5−7	11	6	66	1	11
7−9	8	8	64	9	72
9-11	4	10	40	25	40
Total	50		254		246

$$\bar{x} = \frac{\sum f_i x_i}{n} = \frac{254}{50} = 5.08 \approx 5$$

$$\sigma^2 = \frac{\sum f_i(x_i-\bar{x})^2}{n} = \frac{246}{50} = 4.92$$

$$\sigma = \sqrt{4.92} = 2.22$$

Coefficient of Variation , $CV = \frac{\sigma}{\bar{x}} \times 100 = 45\%$

Advantages of standard deviation:

- Takes all data into account.
- Lends itself to computation of other stable measures.
- Average of deviations around the mean.
- Majority of data within one standard deviation above or below the mean.
- Not expressed in squared units, so makes more sense descriptively

Disadvantages of standard deviation:

- Hard to interpret.
- Can be influenced by extreme scores.

Central tendency and Skewness

Relationship between mean, median and mode may give some indicating of the shape of the frequency distribution without having to create a histogram/ frequency curve.

Skewness: The asymmetric property of frequency distribution is known as skewness.

Measures of skewness based on central tendency:

i) Absolute measures of skewness.
ii) Relative measures of skewness.

i) Absolute measures of skewness:

(a) When

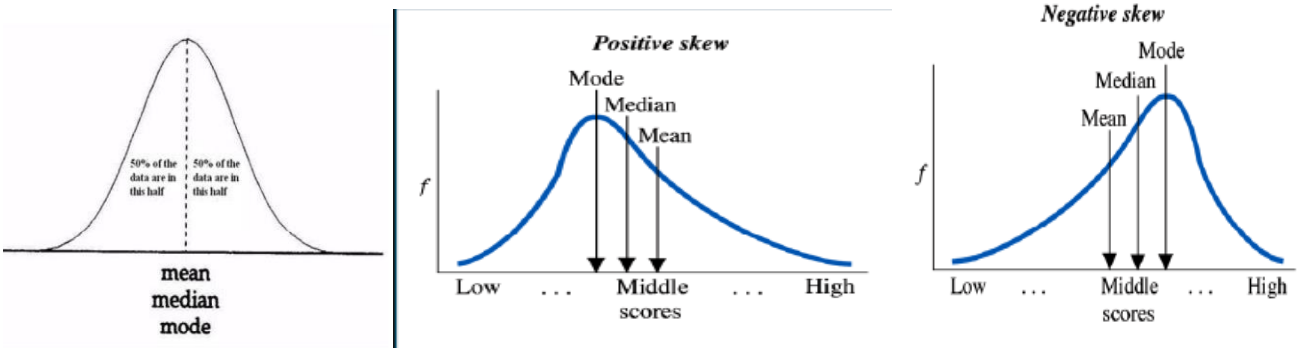
Mean = Median= Mode [the distribution is symmetric]

Mean > Median > Mode [the distribution is positively Skewed]

Mean < Median < Mode [the distribution is negatively Skewed]

(b) Measure of skewness, SK = mean-median if, SK =0[the distribution is symmetric]
Or, = mean – mode SK>0 [the distribution is positively Skewed]
SK<0 [the distribution is negatively Skewed]

Skewness using graph:



Find the mean, median and mode for each set of data:

1. 4, 6, 9, 12, 5; mean = 7.2; median = 6; mode = no mode
2. 7, 13, 4, 7; mean = 7.75; median = 7; mode = 7
3. 10, 3, 8, 15; mean = 9; median = 9; mode = no mode
4. 9, 9, 9, 9, 8; mean = 8.8; median = 9; mode = 9
5. 300, 24, 40, 50, 60; mean = 96.8; median = 50; mode = no mode
6. 23, 23, 12, 12; mean = 17.5; median = 17.5; mode = 12, 23

Comment on the skewness of the above distributions.

MATLAB code for the related topic:

Sum computes the sum of the columns of a matrix

```
>> x=[1 2 3; 4 5 6]

x =

1 2 3

4 5 6
```

- column sum

```
>> sum(x)

ans =

5 7 9
```

- row sum

```
>> sum(x')

ans =

6 15
```

mean(x) computes the mean of a vector or matrix

```
>> x=[1 2 3; 4 5 6]

x =

1 2 327

4 5 6

>> mean(x)

ans =

2.5000 3.5000 4.5000
```

var computes the sample variance of a vector or matrix

```
>> x=[1 2 3; 4 5 6]

x =

1 2 3

4 5 6

>> var(x)

ans =

4.5000 4.5000 4.5000
```

cov computes the sample covariance of a vector or matrix

```
x =

1 2 3

4 5 6

>> cov(x)

ans =

4.5000    4.5000    4.5000

4.5000    4.5000    4.5000
```

std computes the sample standard deviation of a vector or matrix (column-by-column)

```
>> x=[1 2 3; 4 5 6]

x =

1 2 3

4 5 6
```

>> std(x)

ans =

2.1213 2.1213 2.1213

sort orders the values in a vector or the rows of a matrix from smallest to largest

>> x=[1 5 2; 4 3 6]

x =

1 5 2

4 3 6

>> Y=sort(x)

ans =

1 3 2

4 5 6

>> med = **median**(Y)

>> mod = **mode**(x)

skewness computes the sample skewness of a vector or matrix (column-by-column)

>> x=[1 2 3; 4 5 6]

x =

1 2 3

4 5 6

>> skewness(x)

ans =

0 0 0

kurtosis computes the sample kurtosis of a vector or matrix

>> x=[1 2 3; 4 5 6]

x =

1 2 3

4 5 6

>> kurtosis(x)

ans =

1 1 1

Sample questions related to this part:

Exercise:

1. The distribution of miss calls recorded in a mobile set of an organization in different days is shown below:

Class interval of miss calls:		*4-8*	*8-12*	*12-16*	*16-20*	*20-24*	*total*
No. of days,	*f:*	*8*	*24*	*10*	*5*	*3*	*50*

a. *Calculate arithmetic mean, geometric mean and harmonic mean of yield miss calls.*
b. *Find median and mode of miss calls.*
c. *Calculate coefficient of variation.*
d. *Comment on the skewness of the data*

2. Find arithmetic mean, geometric mean, harmonic mean, median and mode of the following observations:

Observations (x): 10, 18, 5, 7, 12, 5

(i) Also calculate mean deviation.

(ii) Read the data using MATLAB and write MATLAB code for mean, median, mode.

3. A train moves 1st 80 km at speed 75 km/h, 2nd 70 km at speed 85 km/h, 3rd 85 km at speed 66 km/h and 4th 55 km at speed 50 km/h, find the average speed throughout the journey.

4. Calculate standard deviation, mean deviation and C.V. of the following observations:

Observations (x): 2, 8, 7, 0, 5, 9, and 12.

Also read the data using MATLAB and write MATLAB code for variance, standard deviation, CV and Skewness.

5. The distribution of diameter (in mm) of a hole drilled in a sheet metal component is shown below:

Class interval of diameter	*:*	*10-12*	*12-14*	*14-16*	*16-18*	*18-20*
Number of holes	*f:*	*8*	*16*	*12*	*10*	*4*

a. *Calculate standard deviation of diameter.*
b. *Find mean deviation about mean.*
c. *Calculate C. V. of the distribution of diameter.*
d. *Calculate AM, GM, HM, mode and median*

.